

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2018 - 2019)

PRE-MEDICAL : NURTURE COURSE**PHASE : I, II & III**

Test Type : COMMON TEST

SET - 1

Test Pattern : NEET(UG)

TEST DATE : 18 - 11 - 2018

Important Instructions

Do not open this Test Booklet until you are asked to do so

1. A seat marked with Reg. No. will be allotted to each student. The student should ensure that he/she occupies the correct seat only.
If any student is found to have occupied the seat of another student, both the students shall be removed from the examination and shall have to accept any other penalty imposed upon them.
2. Duration of Test is 3 Hours and Questions Paper Contains 180 Questions. The Max. Marks are 720.
3. Student can not use log tables and calculators or any other material in the examination hall.
4. Student must abide by the instructions issued during the examination, by the invigilators or the centre incharge.
5. Before attempting the question paper ensure that it contains all the pages and that no question is missing.
6. Each correct answer carries 4 marks, while 1 mark will be deducted for every wrong answer. Guessing of answer is harmful.
7. A candidate has to write his / her answers in the OMR sheet by darkening the appropriate bubble with the help of Blue / Black Ball Point Pen only as the correct answer(s) of the question attempted.
8. Use of Pencil is strictly prohibited.

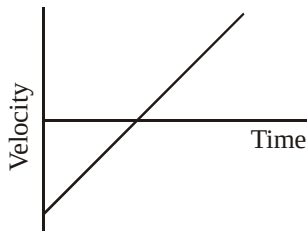
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BEWARE OF NEGATIVE MARKING

1. The graph below shows the velocity with respect to time of an object moving in a straight line. The positive direction is to the right and the negative direction is to the left. Which of the following statements best describes the motion of this object?

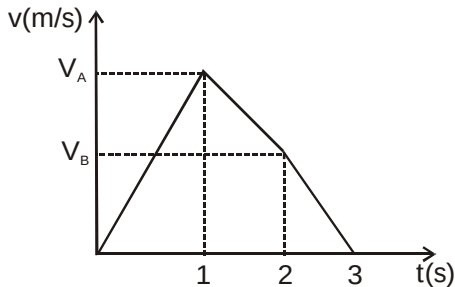


- (1) The object starts at a location to the left of the origin and travels at a constant speed toward the right.
 (2) The object starts at a location to the left of the origin at a slow speed and speeds up as it moves to the right.
 (3) The object slows down as it moves to the left, stops, and starts moving to the right.
 (4) The object slows down as it moves to the right, stops, and continues moving to the right.
2. A car on a highway has an initial speed of 23 m/s. The car accelerates at a constant rate for 10 s to a final speed of 29 m/s. How far does the car travel during this time interval?
 (1) 230 m (2) 260 m
 (3) 275 m (4) 290 m
3. If a particle is moving along straight line and its x-coordinate as a function of time is given below $x = \frac{2}{3}t^3 - 2t^2 + 2t - 5$ then find the time when direction of velocity changes

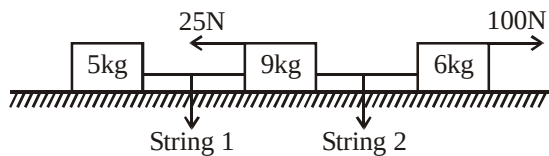
- (1) At $t = 1$ sec
 (2) At $t = 2$ sec
 (3) At $t = 5$ sec
 (4) Direction of velocity remains unchanged
4. A force of 12 N and another force of 5 N can be applied together to produce the effect of a single force. Which of the following force can not be obtained :
 (1) 14 N (2) 8 N
 (3) 18 N (4) 16 N
5. A body of mass 0.5 kg travels in a straight line with velocity $v = ax^{3/2}$ where $a = 5 \text{ m}^{-1/2}\text{s}^{-1}$. The work done by the net force during its displacement from $x = 0$ to $x = 2$ m is
 (1) 1.5 J (2) 50 J
 (3) 10 J (4) 100 J
6. A train is moving slowly on a straight track with a constant speed of 2 ms^{-1} . A passenger in that train starts walking at a steady speed of 2 ms^{-1} with respect to train to the back of the train in the opposite direction of the motion of the train. So to an observer standing on the platform directly in front of that passenger, the velocity of the passenger appears to be :-
 (1) 2 ms^{-1} in the opposite direction of the train
 (2) zero
 (3) 4 ms^{-1}
 (4) 2 ms^{-1}

SPACE FOR ROUGH WORK

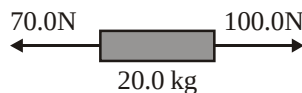
7. The velocity v - s time graph of particle moving on a straight line is shown. The average velocity on the 1st one second is v_1 m/s and the average velocity on the 2nd one second is v_2 m/s. What is the average velocity on the last one second.



- (1) $2v_1$ (2) $v_1 + v_2/2$
(3) $2v_2$ (4) $v_2 - v_1$
8. In the given figure tension at the mid-point of string-2. If mass of string-1 is 6 kg and mass of string-2 is 4 kg.

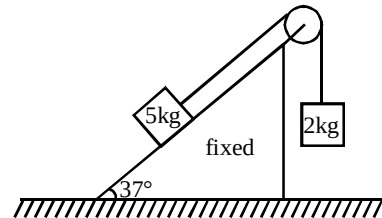


- (1) 70 N (2) 60 N
(3) 80 N (4) 50 N
9. The diagram below shows the horizontal forces on a 20.0 kg mass. The force are constant in time. If the mass starts from rest, how far has it traveled in the horizontal direction after 3.00s ?

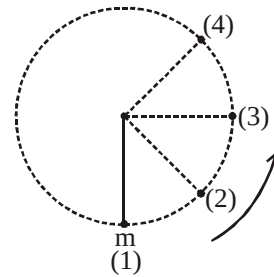


- (1) 4.50 m (2) 6.75 m
(3) 9.00 m (4) 25.5 m

10. In the arrangement shown in figure coefficient of friction between 5kg block and incline plane is $\mu = 0.5$. Friction force acting on the 5kg block is :-



- (1) 20 N (2) 15 N
(3) 10 N (4) 5 N
11. A particle (bob) of mass 1 kg is performing vertical circular motion. Then :



- (1) Power delivered by tension at point (4) is negative.
(2) Power delivered by tension at point (2) is positive.
(3) Power delivered by gravitational force at point (3) is zero
(4) Power delivered by gravitational force at point (4) is negative.

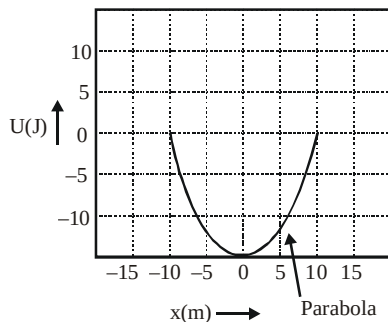
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12. A ball is thrown vertically upwards with a velocity v and an initial kinetic energy E . When half way to the top of its flight, it can have velocity and kinetic energy respectively of :

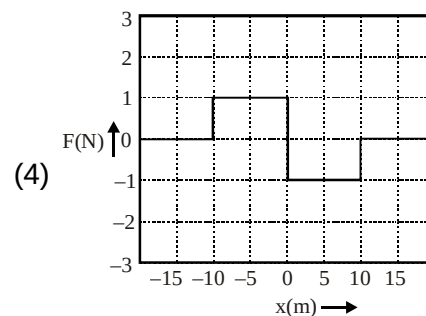
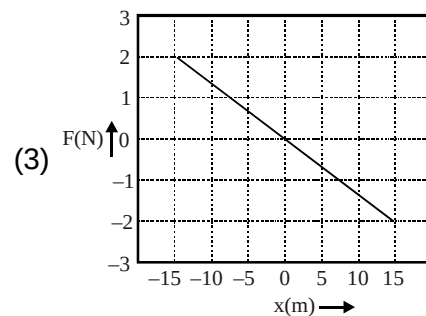
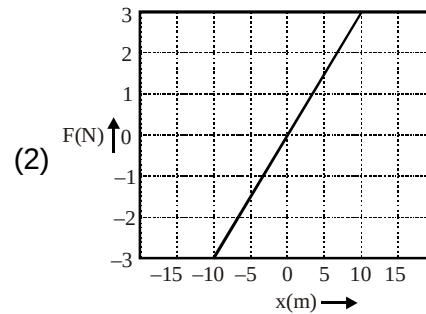
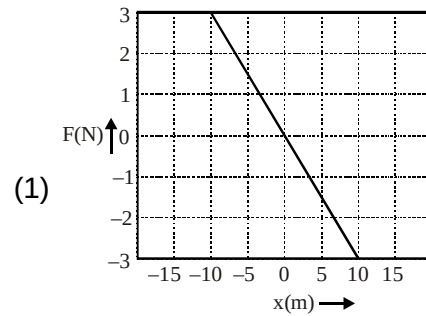
(1) $\frac{v}{2}, \frac{E}{2}$ (2) $\frac{v}{\sqrt{2}}, \frac{E}{2}$

(3) $\frac{v}{4}, \frac{E}{2}$ (4) $\frac{v}{4}, \frac{E}{\sqrt{2}}$

13. Consider the following graph of potential energy (U) vs position (x) for a particle travelling along x -axis. No other contact force exists, as shown in figure.

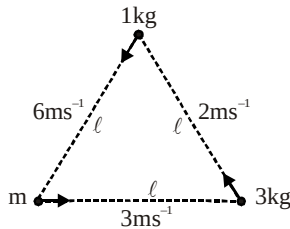


Which of the following graph represents the force corresponding to the given potential ?

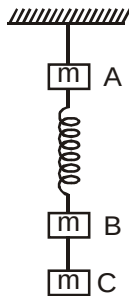


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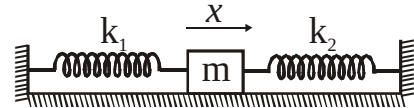
14. Three particles of masses 1kg, m (kg) and 3kg are situated at the corners of an equilateral triangle move at speed 6 ms^{-1} , 3 ms^{-1} and 2 ms^{-1} respectively. Each particle maintains a direction towards the particle at the next corner symmetrically. If the velocity of centre of mass of the system at this instant is zero then find the value of m .



- (1) 1 kg (2) 2 kg
(3) 3 kg (4) 4 kg
15. In the figure shown, find the accelerations of blocks A, B and C immediately after string connecting block A with ceiling is cut :-



- (1) $a_A = 3g\downarrow$, $a_B = g\downarrow$, $a_C = g\downarrow$
(2) $a_A = 3g\downarrow$, $a_B = a_C = 0$
(3) $a_A = 2g\downarrow$, $a_B = a_C = 0$
(4) $a_A = 3g\downarrow$, $a_B = 0$, $a_C = g\downarrow$
16. The block of mass m is displaced by x towards right and released, if there is no friction anywhere, the maximum acceleration produced in the block is

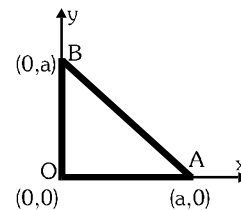


- (1) $\frac{(k_2 + k_1)x}{m}$ (2) $\frac{(k_2 - k_1)x}{m}$
(3) $\frac{k_2 x}{m}$ (4) $\frac{k_1 x}{m}$

17. A deliveryman moves 10 cartons from the sidewalk, along a 10-meter ramp to a loading dock, which is 1.5 meters above the sidewalk. If each carton has a mass of 25 kg, what is the total work done by the deliveryman on the cartons to move them to the loading dock ?



- (1) 2500 J (2) 3750 J
(3) 10000 J (4) 25000 J
18. Three rods of the same mass are placed as shown in the figure. Calculate the coordinates of the centre of mass of the system.



- (1) $\left(\frac{a}{2}, \frac{a}{2}\right)$ (2) $\left(\frac{a}{3}, \frac{a}{2}\right)$
(3) $\left(\frac{a}{2}, \frac{a}{3}\right)$ (4) $\left(\frac{a}{3}, \frac{a}{3}\right)$

SPACE FOR ROUGH WORK

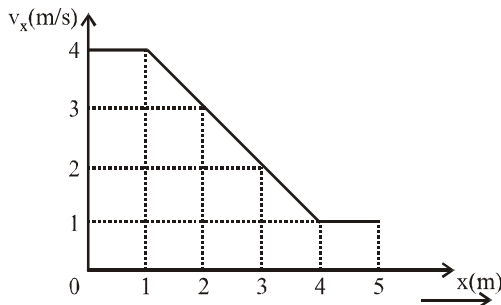
19. A particle is given successive displacements. Which of the following sets of displacements could be capable of returning the particle to its initial position?
- 10 m, 8m, 6 m, 30 m
 - 20 m, 10 m, 6m, 50 m
 - 70m, 20 m, 40 m, 30 m
 - 100 m, 18m, 22 m, 32 m
20. The dimensional formula of product of two physical quantities P and Q is $[ML^2T^{-2}]$ and the dimensional formula of P/Q is $[MT^{-2}]$. P and Q respectively are
- force, velocity
 - work, velocity
 - force, displacement
 - momentum, displacement
21. A man can swim in still water with a speed of 25 m/min. If the speed of the stream is 15 m/min, and its width is 100 m, the respective time taken to cross the stream by shortest route and the quickest route is
- 3 min, 5 min
 - 5 min, 4 min
 - 3 min, 4 min
 - 4 min, 5 min
22. A ball is projected vertically upward from ground at 20 m/s. Immediately after 2 sec what will be the direction of displacement, velocity and acceleration ? ($g = 10 \text{ m/s}^2$)

	Displacement	Velocity	Acceleration
(1)	up	down	down
(2)	down	down	down
(3)	up	down	up
(4)	up	up	down

23. Particle is projected upward with velocity 100 m/s at an angle of 30° with horizontal. Find the time when its velocity is perpendicular to the acceleration :-
- 10 s
 - 5 s
 - 20 s
 - 10 s
24. A student measures the distance traversed in free fall of a body, initially at rest in a given time. He uses this data to estimate g, the acceleration due to gravity. If the maximum percentage errors in measurement of the distance and the time are e_1 and e_2 respectively, the percentage error in the estimation of g is :-
- $e_1 + 2e_2$
 - $e_1 + e_2$
 - $e_1 - 2e_2$
 - $e_2 - e_1$
25. Two persons A and B are sitting in two cars 1 and 2 such that the velocity of the cars is given by $\vec{v}_1 = 3t^2\hat{i} - \frac{4}{3}t^6\hat{j}$ and $\vec{v}_2 = 3(t^2 - 5t)\hat{i}$ respectively. The fictitious (pseudo) force on B as observed by A at $t = 1$ sec will be :-
- ($m_A = 70 \text{ kg}$, $m_B = 50 \text{ kg}$)
- $70(4\hat{j} - 3\hat{i})$
 - $100(4\hat{j} - 3\hat{i})$
 - $50(8\hat{j} - 15\hat{i})$
 - $70(8\hat{j} - 15\hat{i})$

SPACE FOR ROUGH WORK

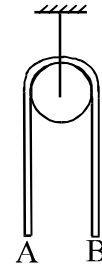
26. A body moves in a straight line for which graph of v_x vs x is shown in the graph. Find the magnitude of acceleration of the body at $x = 3\text{m}$?



- (1) 2 (2) 0
(3) 4 (4) -2
27. In a certain system, the units of length, mass and time are chosen to be 1 cm, 100 g and 10 s. The unit of kinetic energy will be :-
(1) 0.1 erg (2) 1 erg
(3) 10 erg (4) None of these
28. Due to some unknown interaction, force F experienced by a particle is given by the following equation. $\vec{F} = -\frac{A}{r^3} \vec{r}$

Where A is positive constant and r distance of the particle from origin of a coordinate system. Dimensions of constant A are

- (1) ML^2T^{-2} (2) ML^3T^{-2}
(3) ML^4T^{-2} (4) ML^0T^0
29. A uniform chain of length $2L$ is hanging in equilibrium position. If end B is given a slightly downward displacement, the imbalance causes an acceleration. Here pulley is small and smooth & string is inextensible. The acceleration of end B when it has been displaced by distance x , is :-



- (1) $\frac{x}{L}g$ (2) $\frac{2x}{L}g$
(3) $\frac{x}{2}g$ (4) g
30. A stone is projected with speed 20 m/s at angle 37° with the horizontal and it hits the ground with speed 12 m/s due to air resistance. (Assume the effect of air resistance to reduce only horizontal component of velocity). Then the time of flight will be -
(1) greater than 2.4 sec
(2) less than 2.4 sec.
(3) 2.4 sec.
(4) depends on other data
31. If force acting on a particle is $(3\hat{i} - 4\hat{j} + 2\hat{k})\text{N}$ and its velocity vector is $(-2\hat{i} + 2\hat{j} + \hat{k})\text{m/s}$ then magnitude of component of force along velocity is
(1) 1 (2) 2 (3) 3 (4) 4
32. A river is flowing at 3 m/s towards East, a boat is steered toward North at 4 m/s and wind is blowing at 5 m/s 53° North of East. Flag mounted on boat will flutter along
(1) North-East (2) South-West
(3) West (4) Stand still

SPACE FOR ROUGH WORK

33. A toy train in C.B. garden Kota consist of 4 coaches of 100 kg each. Its coaches are joined by a magnetic coupler which can support 1200 N. Mass of engine is 600 Kg. The external horizontal force on engine which is applied by ground for no link to break is (Assume no friction between coaches and rail)

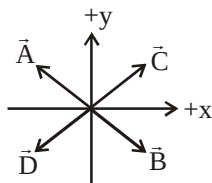
- (1) 1200 N (2) 2400 N
(3) 3000 N (4) 1000 N

34. You have a ring balanced at the center of the table by three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$.

The forces $\vec{F}_1$ and $\vec{F}_2$ have components.

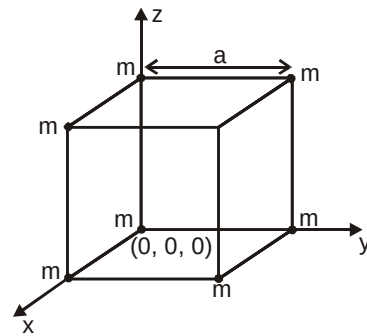
$$F_{1x} = 10 \text{ N} \quad F_{1y} = -50 \text{ N} \quad F_{2x} = -40 \text{ N} \quad F_{2y} = 100 \text{ N}$$

The force $\vec{F}_3$ is required to make the ring stationary at the center of the table. Which one of the 4 vectors in the diagram could represent $\vec{F}_3$?



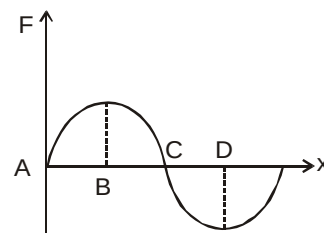
- (1) $\vec{A}$ (2) $\vec{B}$
(3) $\vec{C}$ (4) $\vec{D}$

35. Seven particles, each of mass m are placed at the seven corners of a cube of side ' a ' but one corner is vacant, as shown in figure. The co-ordinates of the centre of mass of the system is :-



- (1) $\left(\frac{2a}{7}, \frac{2a}{7}, \frac{2a}{7}\right)$
(2) $\left(\frac{3a}{7}, \frac{3a}{7}, \frac{3a}{7}\right)$
(3) $\left(\frac{a}{3}, \frac{a}{3}, \frac{a}{3}\right)$
(4) $\left(\frac{5a}{11}, \frac{5a}{11}, \frac{5a}{11}\right)$

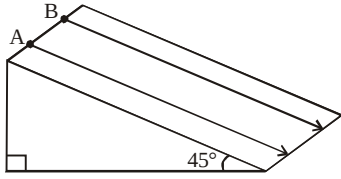
36. Particle of mass '1 kg' is moving on the 'x-axis' such that force on a particle varies with position as shown in graph. Select the position of stable equilibrium for the particle :



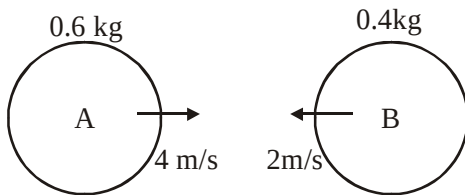
- (1) A (2) B
(3) C (4) D

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37. Two particles A and B are released from the top of incline. If $m_A > m_B$ then :



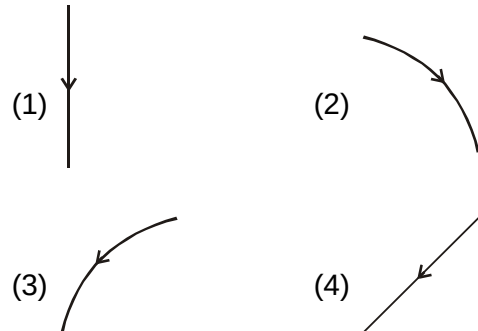
- (1) Both will reach simultaneously if A is released before B
 (2) Both will reach simultaneously if A is released after B
 (3) Both will reach simultaneously if both are released together.
 (4) Both may reach simultaneously if both are released together and A is given some initial velocity.
38. The velocities of two steel balls before impact are shown. If after head on impact the velocity of ball B is observed to be 3 m/s to the right, the coefficient of restitution is



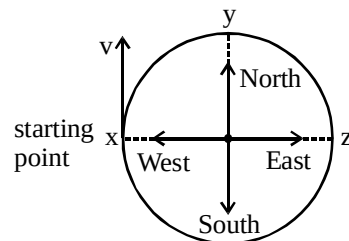
- (1) $\frac{7}{18}$ (2) $\frac{6}{7}$
 (3) $\frac{5}{18}$ (4) None of these
39. A particle of mass 2kg moves in free space such that its position vector varies with time as $\vec{r} = [(3 + 4t^2)\hat{i} + (2t)\hat{j} + (3 - 6t)\hat{k}]m$ where t is in seconds. Net force acting on the particle is

- (1) zero
 (2) parallel to x-axis
 (3) parallel to y-axis
 (4) time dependent

40. A ball is dropped vertically on ground. An observer sitting on a train that moves towards right with constant velocity observes the ball. The trajectory observed is :



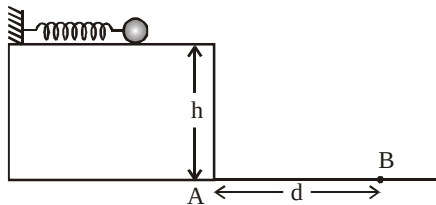
41. A boy is moving with a uniform speed v in a horizontal circle in clockwise direction, starting from x as shown in figure. The change in velocity :



- (1) After first half revolution = $2v$; directed towards south
 (2) After first half revolution = $2v$; directed towards east
 (3) After first quarter revolution = $v\sqrt{2}$; directed towards north-west
 (4) After first quarter revolution = $v\sqrt{2}$; directed towards south-west

SPACE FOR ROUGH WORK

42. If a ball is thrown vertically upwards with speed u , the distance covered during the last 't' seconds of its ascent is :-
- (1) ut
 - (2) $\frac{1}{2}gt^2$
 - (3) $ut - \frac{1}{2}gt^2$
 - (4) $(u + gt)t$
43. A ball of mass m is fired from a spring loaded gun that is mounted on a table. If ball strikes to point B on ground, then find out initial compression of the spring. (Assume table is smooth and spring constant of spring is k) :



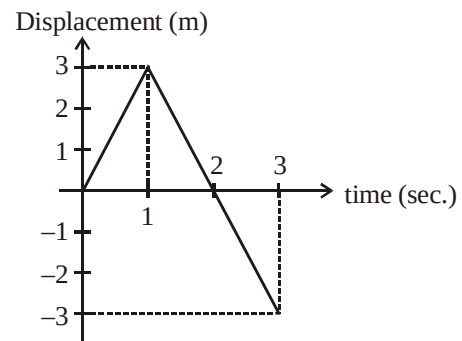
- (1) $\frac{d\sqrt{2h}}{2mg}$
- (2) $\frac{dmg}{2kh}$
- (3) $\frac{d\sqrt{2h}}{mg}$
- (4) $d\sqrt{\frac{mg}{2kh}}$

44. Acceleration of a particle is given by

$$\vec{a} = 6\hat{i} + \frac{5t}{2}\hat{j} \text{ m/s}^2. \text{ At } t = 0 \text{ velocity of the}$$

particle is zero. The magnitude of velocity of the particle at $t = 2$ is :-

- (1) 13 m/s
 - (2) 12 m/s
 - (3) 5 m/s
 - (4) 3 m/s
45. The diagram shows the displacement-time graph for a particle moving in a straight line. Find the average speed for the interval from $t = 0$ to $t = 3$ sec.



- (1) 1.5 m/s
- (2) 3 m/s
- (3) 1 m/s
- (4) 4.5 m/s

SPACE FOR ROUGH WORK

46. Which one of the following statement is incorrect about transition elements ?
 - (1) The last electron enters into them in the d-orbital
 - (2) Their properties are in between those of s-block elements and p-block elements
 - (3) The transition elements with smallest atomic number is scandium
 - (4) Lanthanum is not a transition element but is a lanthanoid
 47. The general configuration d-block elements is :-
 - (1) $ns^2 np^3$
 - (2) $ns^1 (n-1)d^{0-1}$
 - (3) $ns^{0-2} (n-1)d^{1-10}$
 - (4) $ns^{1-2} (n-1)d^{1-10} (n-2)f^3$
 48. Order of shielding effect for same shell :-
 - (1) $s < p < d < f$
 - (2) $s > p > d > f$
 - (3) $p > s > d > f$
 - (4) $s = p = d = f$
 49. Hydrogen has high ionisation energy than alkali metals, due to its :-
 - (1) Large size
 - (2) Small size
 - (3) Ionic bond
 - (4) Covalent bond
 50. Correct increasing order of acidic character is :-
 - (1) $Al_2O_3 < MgO < Na_2O < Cs_2O$
 - (2) $Cs_2O < Na_2O < MgO < Al_2O_3$
 - (3) $Na_2O < MgO < Al_2O_3 < Cs_2O$
 - (4) None of these
 51. Which of the following oxide is neutral :-
 - (1) CO
 - (2) SnO_2
 - (3) SiO_2
 - (4) Na_2O
 52. Correct statements among the following is/are :-
 - (1) 2nd electron gain enthalpy is always endothermic
 - (2) Electronegativity is the property of covalently bonded atoms
 - (3) Al_2O_3 and BeO are amphoteric oxide
 - (4) None of these are incorrect
 53. Unpaired electron is present in :-
 - (1) NO_2^+ and CaO_2
 - (2) KO_2 and $KAlO_2$
 - (3) KO_2 only
 - (4) CaO_2 only
 54. Molecular formula of sodium chloride is :-
 - (1) NaCl
 - (2) $NaCl_2$
 - (3) Na_2Cl
 - (4) None of these
 55. Which of the following compounds do not exist:-
 - (1) PH_5
 - (2) SiF_6^{-2}
 - (3) CF_6^{-2}
 - (4) Both (1) and (3)
 56. The shape of a molecule which has EB_5L_2 molecular formula :-
{E = Central atom, B = Terminal atom, L = lone pair}
 - (1) Square pyramidal
 - (2) Pentagonal planar
 - (3) Octahedral
 - (4) Pentagonal bipyramidal

SPACE FOR ROUGH WORK

57. The shapes of XeF_4 , $\text{XeF}_5^\ominus$ and SnCl_2 are
 (1) Octahedral, trigonal bipyramidal and bent
 (2) Square pyramidal, pentagonal planar and linear
 (3) Square planar, pentagonal planar and angular
 (4) Sea-Saw, T-shaped and linear
58. The species having no $p\pi - p\pi$ bond is -
 (1) ClO_3^- (2) SO_3
 (3) COCl_2 (4) NO_2
59. Which of the following has been arranged in order of decreasing bond length :-
 (1) $\text{P} - \text{O} > \text{Cl} - \text{O} > \text{S} - \text{O}$
 (2) $\text{P} - \text{O} > \text{S} - \text{O} > \text{Cl} - \text{O}$
 (3) $\text{S} - \text{O} > \text{Cl} - \text{O} > \text{P} - \text{O}$
 (4) $\text{Cl} - \text{O} > \text{S} - \text{O} > \text{P} - \text{O}$
60. The correct order of bond angle
 (1) $\text{PF}_3 < \text{PCl}_3 < \text{PBr}_3 < \text{PI}_3$
 (2) $\text{PF}_3 = \text{PCl}_3 = \text{PBr}_3 = \text{PI}_3$
 (3) $\text{PF}_3 > \text{PCl}_3 > \text{PBr}_3 > \text{PI}_3$
 (4) $\text{PF}_3 > \text{PCl}_3 < \text{PBr}_3 > \text{PI}_3$
61. The correct sequence of bond order is:
 (1) $\text{O}_2^- > \text{O}_2^+ > \text{O}_2$
 (2) $\text{O}_2^+ > \text{O}_2^- > \text{O}_2$
 (3) $\text{O}_2 > \text{O}_2^- > \text{O}_2^+$
 (4) $\text{O}_2^+ > \text{O}_2 > \text{O}_2^-$
62. Which of the following angle correspond to sp^2 hybridisation :-
 (1) 90° (2) 120°
 (3) 180° (4) 109°
63. For which of the following molecule significant $\bar{\mu} = 0$

 (a) (b)
 (c) (d) All
 (1) Only (c) (2) (c) and (b)
 (3) Only (a) (4) (a) and (b)
64. The vapour pressure of *o*-nitrophenol at any given temperature is predicted to be-
 (1) higher than that of *p*-nitrophenol
 (2) lower than that of *p*-nitrophenol
 (3) same as that of *p*-nitrophenol
 (4) higher or lower depending upon the size of the vessel
65. Select incorrect order of property :-
 (1) $\text{NaCl} > \text{KCl} > \text{RbCl}$ Covalent character
 (2) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$ Lattice energy
 (3) $\text{Na}_2\text{CO}_3 > \text{K}_2\text{CO}_3 > \text{Rb}_2\text{CO}_3$ Thermal stability
 (4) $\text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O} < \text{HF}$ Thermal stability
66. For true use 'T', For false use 'F' :
 (i) $\text{F}_2 < \text{Cl}_2 < \text{Br}_2 < \text{I}_2$; Boiling point
 (ii) $\text{SiCl}_4 > \text{ZnCl}_2 > \text{CaCl}_2$; Covalent character
 (iii) $\text{P}_x + \text{P}_x$; π -bond inter nuclear axis 'Y'
 (iv) B_2H_6 , AlCl_3 ; e^- deficient
 (1) T, T, T, F (2) T, T, T, T
 (3) F, T, F, T (4) F, T, T, F

SPACE FOR ROUGH WORK

67. The number of oxygen atoms present in 20.4 g of Al_2O_3 are equal to the number of :-
 (1) Oxygen atoms in 3.6 g of water
 (2) Oxygen atoms in 5.4 g of water
 (3) Hydrogen atoms in 5.4 g of water
 (4) Hydrogen atoms in 10.8 g of water
68. What will be the concentration of Pb^{+2} ions when PbSO_4 ($K_{\text{sp}} = 1.8 \times 10^{-8}$) starts to precipitate from a solution of SO_4^{-2} ions with 0.0045 M concentration?
 (1) 4×10^{-6} M
 (2) 4×10^{-8} M
 (3) 2×10^{-8} M
 (4) 8.1×10^{-7} M
69. Which of the following specie can act as Bronsted acid as well as Bronsted base ?
 (1) H_2SO_4 (2) $\text{HCO}_3^\ominus$
 (3) K_2CO_3 (4) HPO_3^{-2}
70. 10 mol SO_2 and 15 mol O_2 were allowed to react the remain moles of SO_2 and O_2 respectively are:

$$2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$$

 (1) 0 mol, 10 mol
 (2) 2 mol, 8 mol
 (3) 4 mol, 5 mol
 (4) 8 mol, 2 mol
71. Find out the radius of first excited state of H-atom
 (1) $0.529 \text{ \AA}$
 (2) $2.12 \text{ \AA}$
 (3) $0.529 \times 2 \text{ \AA}$
 (4) $8.21 \text{ \AA}$
72. Calculate uncertainty in velocity of a particle of mass 2 Kg having uncertainty in position is equal to 100 $\text{\AA}$.
 (1) $2.63 \times 10^{-27} \text{ m s}^{-1}$
 (2) $3.73 \times 10^{-22} \text{ m s}^{-1}$
 (3) $8.32 \times 10^{-25} \text{ m s}^{-1}$
 (4) $4.21 \times 10^{-30} \text{ m s}^{-1}$
73. Velocity of 2nd shell electron of Be^{+++} is :-
 (1) $2.18 \times 10^6 \text{ m/sec}$
 (2) $4.36 \times 10^6 \text{ m/sec}$
 (3) $1.09 \times 10^6 \text{ m/sec}$
 (4) None
74. Wavelength of third line of Balmer series of H is:-
 (1) $\frac{21}{100R}$ (2) $\frac{100}{21R}$
 (3) $\frac{21R}{100}$ (4) $\frac{100R}{21}$
75. Solubility of $\text{Th}_3(\text{PO}_4)_4$ ($M_w = m$) is w g/100 ml at 25°C . It's K_{sp} is :-
 (1) $6.912 \times 10^{10} \left(\frac{w}{m}\right)^7$
 (2) $6.912 \times 10^3 \left(\frac{w}{m}\right)^7$
 (3) $1.08 \times 10^7 \left(\frac{w}{m}\right)^5$
 (4) $10^2 \times \left(\frac{w}{m}\right)^2$

SPACE FOR ROUGH WORK

76. Maximum efficiency of anionic hydrolysis will be shown by :

- (1) CN^- (2) Cl^-
(3) NO_3^- (4) SO_4^{2-}

77. Formula $\text{pK}_h - \text{pK}_w + \text{pK}_b = 0$ is not appropriate for the following salt :-

- (1) CuSO_4 (2) NH_4Cl
(3) AgCl (4) NaCN

78. In any subshell, the maximum number of electrons having same value of spin quantum number is :-

- (1) $\sqrt{l(l+1)}$ (2) $l + 2$
(3) $2l + 1$ (4) $4l + 2$

79. Which of the following subshell can accommodate as many as 10 electrons ?

- (1) 2d (2) 3d
(3) $3d_{xy}$ (4) $3d_{z^2}$

80. For reaction $2\text{X}(\text{g}) + \text{Y}(\text{g}) \rightleftharpoons 2\text{Z}(\text{g})$ value of equilibrium constant (K_c) is 2.25, if at equilibrium concentrations of X and Z are 2 mol L^{-1} and 3 mol L^{-1} respectively then concentration of Y at equilibrium is -

- (1) 1.0 mol L^{-1}
(2) 2.25 mol L^{-1}
(3) 2.0 mol L^{-1}
(4) 5.0 mol L^{-1}

81. K_p for the reaction

$\text{A}(\text{g}) + 2\text{B}(\text{g}) \rightleftharpoons 3\text{C}(\text{g}) + \text{D}(\text{g})$; is 0.01 atm .
What will be its K_c at 1000 K in terms of R ?

(1) $\frac{1.0 \times 10^{-5}}{R}$

(2) $\frac{R}{5 \times 10^{-5}}$

(3) $5 \times 10^{-5}R$

(4) None of these

82.

	Column-I		Column-II
A	$\text{Fe}(\text{NO}_3)_2(\text{aq.})$	P	Only cationic hydrolysis
B	$\text{KClO}_4(\text{aq.})$	Q	Only anionic hydrolysis
C	$\text{HCOONa}(\text{aq.})$	R	both cationic as well as anionic hydrolysis
D	$\text{NH}_4\text{CN}(\text{aq.})$	S	No hydrolysis

(1) A-P, B-S, C-Q, D-R

(2) A-P, B-S, C-R, D-Q

(3) A-P, B-Q, C-S, D-R

(4) A-P, B-R, C-S, D-Q

83. The K_a values for HPO_4^{2-} and HSO_3^- are 4.8×10^{-13} and 6.3×10^{-8} respectively. Therefore, it follows the HPO_4^{2-} is a acid than HSO_3^- and PO_4^{3-} is a base than SO_3^{2-}

(1) weaker, stronger

(2) stronger, weaker

(3) weaker, weaker

(4) stronger, stronger

SPACE FOR ROUGH WORK

84. 0.6 mole of PCl_5 , 0.3 mole of PCl_3 and 0.5 mole of Cl_2 are taken in a 1 L flask to obtain the following equilibrium :



If the equilibrium constant K_c for the reaction is 0.2. Predict the direction of the reaction.

- (1) Forward direction
- (2) Backward direction
- (3) Direction of the reaction cannot be predicted
- (4) Reaction does not move in any direction

85. If the equilibrium constant is K_c in the reaction $2\text{NO}(g) \rightleftharpoons \text{N}_2(g) + \text{O}_2(g)$, the equilibrium concentration of NO will be :-

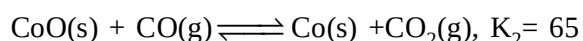
$$(1) \frac{[\text{N}_2]}{[\text{K}_c]^{1/2}}$$

$$(2) \frac{[\text{N}_2][\text{O}_2]^{1/2}}{K_c}$$

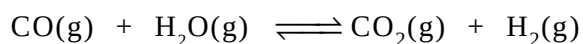
$$(3) [\text{O}_2][\text{K}_c]^{-1/2}$$

(4) All of these

86. The equilibrium constant for the reaction at 1000K are -



The equilibrium constant for the reaction,



at 1000K is -

- (1) 0.13
- (2) 7.69
- (3) 0.0179
- (4) 69.3

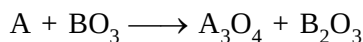
87. The mass of 12.04×10^{21} molecules of O_3 will be:-

- (1) 96g
- (2) 9.6g
- (3) 0.96g
- (4) 960g

88. Number of atoms in 560 g of Fe is (atomic mass = 56 gm/mol) :-

- (1) is twice that of 70g N atoms
- (2) is half that of 20g H atoms
- (3) is same as 8g oxygen atom
- (4) (1) and (2) both

89. According to following reaction :



The number of moles of A_3O_4 produced if 1 mole of A is mixed with 1 mole of BO_3 .

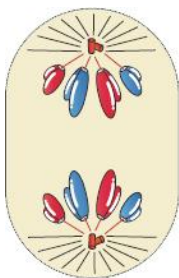
- (1) 3
- (2) 1
- (3) 1/3
- (4) 2/3

90. pH of 10^{-7}M aqueous solution of $\text{Ba}(\text{OH})_2$ at 25°C is :

- (1) 6.52
- (2) 7.48
- (3) 7
- (4) 6.3

SPACE FOR ROUGH WORK

91. Identify the following figure of cell cycle :-



- (1) Anaphase I (2) Metaphase I
(3) Anaphase II (4) Telophase II

92. 'x' shaped structure chiasmata appear in which stage of cell cycle ?

- (1) Prophase I of meiosis I
(2) Anaphase I of meiosis I
(3) Anaphase of mitosis
(4) Prophase of mitosis

93. Which of the following is **correct** event for G_2 phase of interphase ?

- (1) DNA replication in the nucleus and the centriole duplication in the cytoplasm
(2) Attachment of spindle fibres to kinetochores of chromosomes
(3) Pairing of homologous chromosomes
(4) Proteins are synthesised in preparation for mitosis

94. In which division the homologous chromosomes pair to form bivalents and undergo crossing over:-

- (1) Mitosis (2) Meiosis I
(3) Meiosis II (4) Fission

95. Which one of the following is **not** a feature of ribosomes ?

- (1) Composed of RNA and protein
(2) Not surrounded by any membrane
(3) Found within four cell organelles
(4) Main site for protein synthesis

96. Several ribosomes attach to a single m-RNA and form a chain called as :-

- (1) Polysome (2) Promotor
(3) Axoneme (4) Inclusion bodies

97. Which cell organelles help in oxidative phosphorylation and generation of adenosine triphosphate :-

- (1) ER (2) Golgi body
(3) Chromoplast (4) Mitochondria

98. Which of the following cellular structure contain pigments ?

- (1) ER, Golgi body, Chloroplast
(2) Vacuole, Chromoplast, Chloroplast
(3) Chloroplast, Chromoplast, Leucoplast
(4) Chloroplast, Nucleus, Mitochondria

99. In which stage of cell cycle chromatin are full condensed :-

- (1) Interphase (2) Prophase
(3) Telophase (4) Metaphase

100. Which of the following statement is **not correct** with S phase of cell cycle ?

- (1) Chromosome number double but DNA content is same
(2) Centriole duplication in cytoplasm of animal cell
(3) Formation of histone protein
(4) DNA replication in nucleoplasm of animal cell

101. Three of the following statements are correct while one is incorrect. Find out the **incorrect** statement:-

- (1) Electron microscopic study of a cilium show that they are covered with plasma membrane
(2) The golgi apparatus principally performs the function of packaging of material
(3) Cell membrane is composed of lipids that are arranged in a bilayer and also possess protein and carbohydrates
(4) During anaphase centrioles are move to opposite pole

102. Which of the following option is **not** related with mitochondria ?

- (1) Photophosphorylation
(2) Formation of ATP
(3) Endosymbiotic origin
(4) Division by fission

103. Match the column I with column II :-

	Column I		Column II
I	Primary constriction	a	Gives the appearance of small fragement called satellite
II	Secondary constriction	b	Beads on string structure in chromatin
III	Telomere	c	Kinetochores are present
IV	Nucleosome	d	Related with ageing process

- (1) I-c, II-d, III-b, IV-a (2) I-b, II-c, III-d, IV-a
(3) I-c, II-a, III-d, IV-b (4) I-c, II-b, III-a, IV-d

104. Which one of the following options gives the correct events of cell cycle according to phases of cell division A, B and C :-

	A Telophase of mitosis	B Anaphase I of meiosis	C Prophase of mitosis
(1)	ER, Golgi body reform	Sister chromatids separate	Nuclear membrane disappear
(2)	Nuclear membrane reform	Centromere divide	Chromosome move to pole
(3)	Chromosomes lost their individuality	Homologous chromosomes separate	Centriole move to pole
(4)	ER, Golgi body reform	Chromosome move to pole	Chromosome move to equator

105. The osmotic expansion of a cell kept in water is chiefly regulated by :-

- (1) Mitochondria (2) Vacuole
(3) Plastids (4) Ribosomes

106. During which phase (s) of cell cycle amount of DNA in a cell at 4C, if the initial amount is denoted as 2C :-

- (1) G_0 and S (2) G_1 and S
(3) G_2 (4) G_1 and M

107. Which one of the following living organisms completely lacks a cell wall ?

- (1) Cyanobacteria (2) *Amoeba*
(3) Potato (4) Pea

108. Consider the following four statements (a-d) and select the option which includes all the **correct** ones only :-

- (a) Ribosomes are the granular structure first observed by George Palade.
(b) Both the centrioles in a centrosome lie perpendicular to each other.
(c) A single human cell has approximately 2.2 meter long thread of DNA distributed in one chromosomes.
(d) Material of the nucleus (chromatin) stained by the only acidic dyes.

- (1) a, b and c (2) a, c and d
(3) a and b (4) a, b and d

109. Consider the following four statements (a-d) and select the option which includes all the correct ones only :-

- (a) Na^+ pump is an example of active transport.
(b) All living organisms are composed of cells so that all cells have similar shape.
(c) The cytoplasm is the main area of cellular activity.
(d) At present time cells are formed *de novo* from abiotic materials.

Options :-

- (1) Statements a, b and c (2) Statements a, c and d
(3) Statements b and d (4) Statements a and c

110. As the polar molecules cannot pass through the non polar I, they require II of the membrane to facilitate their transport across the membrane. Choose the **correct** word for I and II :-

	I	II
(1)	Cholesterol	Carbohydrate
(2)	Protein	Lipid
(3)	Carbohydrate	Protein
(4)	Lipid bilayer	Protein

111. Select the **correct** statement :-

- (1) Histone protein is present in all prokaryotes but absent in animal cell
- (2) ER is absent in eukaryotes but present in prokaryotes
- (3) Nuclear membrane is absent in prokaryotes but nuclear membrane is present in eukaryotes
- (4) Ribosome is present in eukaryotic cells but absent in prokaryotic cells

112. The fluid nature of the membrane is important from the point of various function. Which one of the following is **not** the function of fluidity:-

- (1) Cell growth
- (2) Translation
- (3) Endocytosis
- (4) Formation of intercellular junction

113. An improved model of the cell membrane was proposed by :-

- (1) Singer and Nicolson
- (2) Watson and Crick
- (3) A.V. Leeuwenhoek
- (4) Davson and Danielli

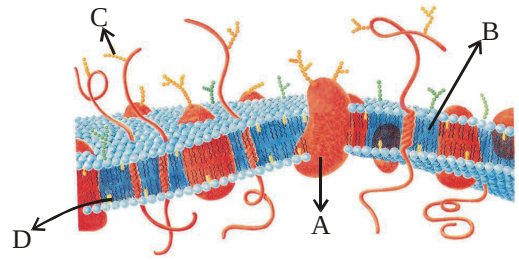
114. Which one of the following is **wrong** statements?

- (1) Anything less than a complete structural of a cell does not ensure independent living
- (2) In eukaryotic cell there is an extensive compartmentalisation of cytoplasm through the presence of membrane less cell organelles
- (3) One of the most important function of the plasma membrane is the transport of molecules across it
- (4) In active transport ATP is utilised

115. Spectrin of erythrocytes membrane can be easily dissociated by high ionic strength and metal ion chelating agent are example of :-

- (1) Intrinsic protein (2) Channel protein
- (3) Peripheral protein (4) Cytoplasmic protein

116. Given below in the figure of cell membrane. Identify the labelled part (A,B,C,D) with their respective statement :-



	Labelled part	Statement
(1)	B – intrinsic protein	Help in cell to cell recognition
(2)	A – Lipid bilayer	Absent in prokaryotic cell
(3)	C – Sugar	Percentage is minimum in erythrocyte membrane
(4)	D – Glycoprotein	Provide fluidity to cell membrane

117. The basal body of cilia or flagella are structurally equivalent to :-

- (1) Centriole
- (2) Kinetochore
- (3) Centromere
- (4) Nuclear matrix

118. (A) Present in prokaryotes and eukaryotes
(B) Composed of ribonucleic acid and proteins
Above statements are true for :-

- (1) Vacuoles
- (2) Chloroplast
- (3) Mesosome
- (4) Ribosome

119. The major site for the synthesis of lipid in a eukaryotic cell is :-

- (1) Golgi apparatus
- (2) Smooth endoplasmic reticulum
- (3) Microbodies
- (4) Centrosome

120. "Chromatin" name was given by ?

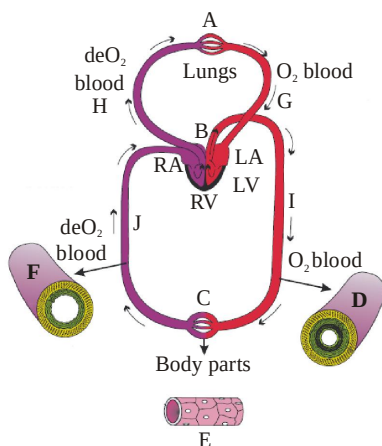
- (1) Starsburger
- (2) Schleiden
- (3) Robert brown
- (4) Flemming

121. Which of the following statement is incorrect?
 (1) Succus entericus is the name given to intestinal juice.
 (2) In terrestrial vertebrates lungs are used for the exchange of gases.
 (3) Blood group O^+ is called universal donor.
 (4) Gills perform respiratory function in crustaceans.
122. Match the following columns and choose the correct answer from the options given below :-

Column-A	Column-B	Column-C
a. Uricotelism	i. Aquatic amphibians	u. two ventricles in heart
b. Ureotelism	ii. Elasmobranch fishes	v. two atria in heart
c. Ammonotelism	iii. Birds	w. one ventricle in heart

- (1) a, i, u (2) c, i, u
 (3) b, ii, w (4) a, iii, w

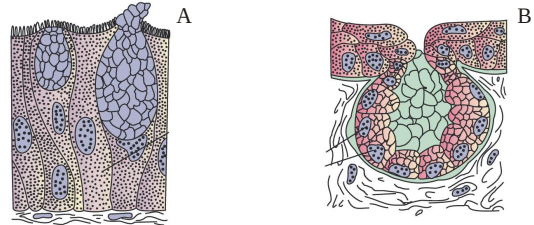
123.



Which of the option is incorrect regarding the above given diagram ?

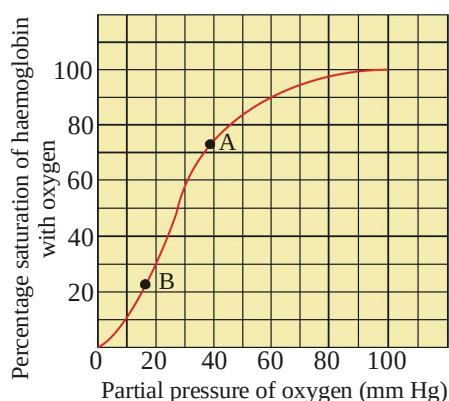
- (1) At 'A' pO_2 is 104 mmHg, pCO_2 is 40 mmHg
 (2) In 'B' and 'E' epithelium is simple squamous epithelium
 (3) 'G' is a vein which contains maximum amount of urea
 (4) 'F' is less muscular and has valves.
124. Which of the following organ/s in human body receive oxygenated as well as deoxygenated blood ?
 (1) liver and kidney
 (2) lungs and kidney
 (3) liver and heart
 (4) heart and kidney

125. The two sketches 'A' and 'B' given below, represent different types of glands. Which one of these is correctly identified in the options given ?



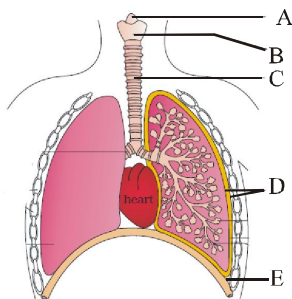
- (1) A-unicellular gland-e.g.-goblet cells secrete mucus which protect the simple columnar glandular epithelium of stomach from excoriation of concentrated HCl
 (2) A-unicellular gland-e.g.- goblet cells secrete mucus which protect the serosa of stomach from excoriation of concentrated HCl
 (3) B-multicellular gland e.g.-goblet cells secrete mucus which protect inner lining of stomach from exocriation of concentrated HCl
 (4) A-unicellular gland-e.g.-goblet cells secrete mucus which protect inner lining of stomach from excoriation of dilute HCl
126. Epithelial cells have a variety of junctions between them. Why ?
 (1) because they have rich supply of blood vessels.
 (2) because cells are loosely packed with each other
 (3) because they provide structural & functional link
 (4) because cells are flat or cuboidal or columnar in shape
127. Choose the correctly matched pairs :-
 (1) Human heart → Myogenic
 (2) Adipose tissue → storage of carbohydrates
 (3) Cuboidal epithelium → Bowman's capsule of nephron
 (4) PCT of nephron → simple columnar epithelium
128. Air from the nasal chambers moves immediately into _____ during inspiration ?
 (1) Alveoli (2) Nasopharynx
 (3) Trachea (4) Larynx
129. Volume of thoracic cavity is increased in –
 (1) Peristalsis (2) Expiration
 (3) Inspiration (4) All of these
130. Volume of air inspired during normal inspiration is ?
 (1) 500 ml (2) 1000 ml
 (3) 3500 ml (4) 1200 ml

131. Higher $p\text{CO}_2$ is found in :-
 (1) Atmosphere (2) Oxygenated blood
 (3) Deoxygenated blood (4) Alveoli
132. Building blocks of our body are :-
 (1) Fats (2) Vitamins
 (3) Carbohydrate (4) Proteins
133. Different types of respiratory organs are found in :-
 (1) Aves (2) Arthropods
 (3) Mammals (4) Sponges
134. Observe the oxyhaemoglobin dissociation curve :-



Now find out the correct option :-

- (1) At 'B' $p\text{O}_2 = 95$ mm Hg
 (2) At 'A' $p\text{O}_2 = 40$ mm Hg
 (3) At 'A' $p\text{CO}_2 = 95$ mm Hg
 (4) At 'A' $p\text{CO}_2 = 104$ mm Hg
135. Mast cells are associated with :-
 (1) exocrine glands
 (2) endocrine glands
 (3) areolar connective tissue
 (4) muscular tissue
136. Identify the correct statement related to the diagram given below :-



- (1) 'B' is a bony box like structure which help in sound production
 (2) 'A' prevent food to enter into food pipe
 (3) 'C' is a long tube with ventrally incomplete C-shaped cartilagenous rings
 (4) Wall of 'D' is double layered with fluid between them

137. Enzyme chymotrypsinogen is secreted by :-
 (1) intestine (2) pancreas
 (3) stomach (4) buccal cavity
138. Number of salivary glands in human beings are:-
 (1) 2 pairs (2) 3 pairs
 (3) 4 pairs (4) 5 pairs
139. Type of dentition in human is :-
 (1) Heterodont and diphyodont
 (2) Heterodont and pleurodont
 (3) Thecodont and homodont
 (4) Thecodont and pleurodont
140. Muscular sphincter which regulates the opening of oesophagus into the stomach is ?
 (1) Pyloric sphincter
 (2) Gastro-oesophageal sphincter
 (3) Gastric valve
 (4) Ileo-Caecal valve
141. Opening of common hepato-pancreatic duct into duodenum is guarded by :-
 (1) Sphincter of Boyden (2) Cardiac sphincter
 (3) Pyloric sphincter (4) Sphincter of Oddi
142. The endocrine part of pancreas secretes :-
 (1) Hormones-insulin and glucagon
 (2) Alkaline pancreatic juice containing hormones
 (3) Alkaline pancreatic juice containing enzymes
 (4) Acidic pancreatic juice containing enzymes
143. Function of gall bladder is :-
 (1) Storage of bile juice
 (2) Formation of enzymes
 (3) Synthesis of bile
 (4) Formation of bile salts
144. Digestive enzymes responsible for breaking down disaccharides includes :-
 (1) Pepsin, Trypsin (2) Lactase, Lipase
 (3) Maltase, Lactase (4) Chymotrypsin, pepsin
145. Which of the following is not the function of HCl in stomach ?
 (1) Break down of proteins into peptones
 (2) Killing the bacteria ingested with food and drinks
 (3) Softening food particles
 (4) Convert inactive protein digesting enzymes into active protein digesting enzymes
146. Which of the following is a protein-energy malnutrition related disorder ?
 (1) Kwashiorkar (2) Marasmus
 (3) Both (1) and (2) (4) Xerophthalmia

- 147.** Infection in alimentary canal is not caused by which of the following organism ?
 (1) Tapeworm (2) Roundworm
 (3) Lactobacilli (4) Hookworm
- 148.** Why do the eyes of patient turns yellow during jaundice ?
 (1) Due to the deposition of bile pigment
 (2) Due to ejection of stomach content through mouth
 (3) Due to stomach malfunctioning
 (4) Due to excessive vomiting
- 149.** Which of the following tissue performs the function of linking and supporting other tissues of the body?
 (1) Epithelial tissue (2) Muscular tissue
 (3) Connective tissue (4) Nervous tissue
- 150.** Select the correctly matched pair from following:-
 (1) Tendon : Dense irregular connective tissue
 (2) Adipose tissue : Loose connective tissue
 (3) Bone : Loose connective tissue
 (4) Areolar tissue : Specialized connective tissue
- 151.** Select the incorrect statement from the following-
 (1) Muscles play an active role in all the movements of the body.
 (2) Blood is the main circulating fluid that helps in the transport of various substances.
 (3) Compound epithelium provides protection against chemical and mechanical stress.
 (4) Adipose tissue serves as a support framework for epithelium
- 152.** Given below the list of some cells that are found in different tissues.
 Fibroblast, Osteocytes, Mast cells, Red blood cells, chondrocytes, Macrophages cells, Platelets.
 How many cells are found mainly in areolar connective tissue ?
 (1) four (2) three
 (3) five (4) All
- 153.** Find out incorrect statement
 (1) Atria and ventricles are separated by a thick muscular septum
 (2) Right atrium opens in to right ventricle through tricuspid valve
 (3) Pericardium is double layered membrane
 (4) Heart is mesodermal in origin
- 154.** Correct reason of entering of blood into atria is:
 (1) Opening of oblique septum
 (2) Closing of bicuspid valve
 (3) Pressure difference between large veins and atria
 (4) Opening of semilunar valve
- 155.** Wind pipe of humans divide into bronchi at the level of -
 (1) 3rd cervical vertebra
 (2) 5th Lumbar vertebra
 (3) 5th thoracic vertebra
 (4) 3rd thoracic vertebra
- 156.** In ECG, P-wave refers to -
 (1) End of ventricular systole
 (2) End of atrial systole
 (3) Atrial depolarisation
 (4) Ventricular depolarisation
- 157.** Which of the following does not shift the oxyhaemoglobin dissociation curve to the right?
 (1) Increased pH
 (2) Increased CO₂
 (3) Increased temperature
 (4) Increased H⁺ concentration
- 158.** Contraction of diaphragm _____ volume of the thoracic cavity in _____
 (1) Increases, antero-posterior axis
 (2) Increases, Dorso-ventral axis
 (3) Decreases, antero-posterior axis
 (4) Decreases, Dorso-ventral axis
- 159.** In alveolar air, the partial pressure of CO₂ is -
 (1) 42 mm Hg (2) 44 mm Hg
 (3) 40 mm Hg (4) 45 mm Hg

- 160.** The carbon dioxide partial pressure in the lung capillaries when blood enters in :-
 (1) Less than that in alveolar air
 (2) More than that in alveolar air
 (3) Equal to that in alveolar air
 (4) Similar to oxygen pressure in the pulmonary capillary
- 161.** Which epithelium is found in wall of blood vessels and air sacs of lungs?
 (1) Simple squamous (2) Simple cuboidal
 (3) Simple columnar (4) Pseudostratified
- 162.** Which epithelium is germinal epithelium ?
 (1) Pseudo stratified (2) Simple columnar
 (3) Simple squamous (4) Simple cuboidal
- 163.** Lining of stomach and intestine helps in absorption and secretion is -
 (1) Simple cuboidal
 (2) Simple columnar
 (3) Simple squamous
 (4) Pseudo stratified
- 164.** When heart rate increases than normal, the condition is called :-
 (1) Bradycardia (2) Cardiac arrest
 (3) Tachycardia (4) Heart failure
- 165.** Blood pressure is measured by which instrument?
 (1) Stethoscope (2) Barometer
 (3) Sphygmomanometer (4) Thermometer
- 166.** Identify the correct value of blood pressure which show hypotension :-
 (1) 100/70 (2) 130/80
 (3) 120/80 (4) 210/140
- 167.** To obtain a standard ECG, a patient is connected to the machine with leads at -
 (1) Both wrists and left ankle
 (2) Left wrist and both ankle
 (3) Both wrists and both ankle
 (4) Both wrists and right ankle
- 168.** Which blood protein play a major role in osmotic balance ?
 (1) Albumin (2) Globulin
 (3) Fibrinogen (4) All of these
- 169.** During diastole auricles receive :-
 (1) Oxygenated blood
 (2) Deoxygenated blood
 (3) Mixed blood
 (4) Venous blood
- 170.** Which of the following has no muscular wall ?
 (1) Capillary (2) Arteriole
 (3) Vein (4) Artery
- 171.** The middle man of the body is :-
 (1) Blood (2) Plasma
 (3) Lymph (4) Serum
- 172.** Which cell of connective tissue, produce and secrete fibres ?
 (1) Mast cell (2) Fibroblast
 (3) Macrophage (4) Adipocytes
- 173.** Which tissue is most abundant and widely distributed in the body of complex animals ?
 (1) Epithelium Tissue (2) Connective tissue
 (3) Muscular tissue (4) Nervous tissue
- 174.** The type of epithelial tissue which line the inner surface of fallopian tubes, smaller bronchi and bronchioles
 (1) Cubical epithelium
 (2) Ciliated epithelium
 (3) Squamous epithelium
 (4) Columnar epithelium
- 175.** The intercellular material of the given figure is solid and resist compression . Identify the figure and the label marked as A and B.

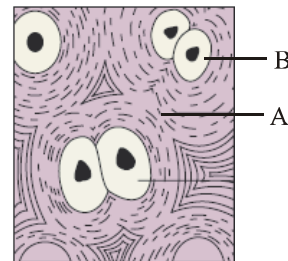


	Figure	A	B
1	Cartilage	Collagen	Chondrocyte
2	Cartilage	Collagen	Chondroclast
3	Bone	Mitochondria	Chondroclast
4	Bone	Collagen fibres	Osteoblast

- 176.** Statement A : cartilage and bone are rigid connective tissue.
Statement B : Blood is the connective tissue in which plasma is the matrix.
(1) Both the statements are false
(2) Statement A is true and statement B is false.
(3) Statement A and statement B are true but statement B is not the correct explanation for statement A
(4) Statement A and statement B are true but statement B is the correct explanation for statement A
- 177.** The successive waves of muscular contractions by which the bolus further passes down through the oesophagus is called :-
(1) Deglutition (2) Churning movement
(3) Masticalion (4) Peristalsis
- 178.** 30% starch is hydrolysed in Buccal cavity by which of the following enzyme ?
(1) Pancreatic α -amylase
(2) Lipase
(3) Maltase
(4) Salivary amylase
- 179.** Which of the following enzymes is not found in the pancreatic juice ?
(1) Amylases (2) Lipases
(3) Enterokinase (4) Trypsinogen
- 180.** Which of the following help in protection of gastric mucosal epithelium from excoriation by HCl ?
(1) Castle's intrinsic factor
(2) Mucous and bicarbonates
(3) Mucous and pepsin
(4) Pepsin and Rennin

